Algebra: Graphing Equations by Plotting Points

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Name: ______________________________ Date: ______________________________ Score: ______________________

DIRECTIONS *Complete the table of values for each equation, then identify the type of graph (line, parabola, etc.).*

1 Find y when x=3:

$$y = 2x - 1$$

Answer: ___________

2 Find y when x=-2:

$$y = 3x + 4$$

Answer: ___________

3 Find y when x=2:

$$y = x^2 - 3$$

Answer: ___________

4 Find y when x=-1:

$$y = x^2 + 2x$$

Answer: ___________

5 Find y when x=0:

$$y = -2x + 5$$

Answer: ___________

6 Find y when x=4:

$$y = (1/2)x - 1$$

Answer: ___________

7 Find x-intercept:

$$y = 2x - 6$$

Answer: ___________

8 Find y-intercept:

$$y = 3x - 9$$

Answer: ___________

9 Find y when x=-3:

$$y = x^2 - x + 1$$

Answer: ___________

10 Identify graph type:

$$y = x^2 + 2$$

Answer: ___________

Answer Key & Solutions

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TEACHER NOTES *Choose easy x-values (often -2, -1, 0, 1, 2). Substitute each into the equation and solve for y. Plot and connect the po ints.*

1 y=2x-1, x=3

= y = 5

y=2(3)-1=6-1=5

2 y=3x+4, x=-2

= y = -2

y=3(-2)+4=-6+4=-2

3 y=x^2-3, x=2

= y = 1

y=4-3=1

4 y=x^2+2x, x=-1

= y = -1

y=1-2=-1

5 y=-2x+5, x=0

= y = 5

y-intercept: y=5

6 y=1/2x-1, x=4

= y = 1

y=2-1=1

7 y=2x-6 x-intercept

= x = 3

Set y=0: 0=2x-6, x=3

8 y=3x-9 y-intercept

= y = -9

Set x=0: y=-9

9 y=x^2-x+1, x=-3

= y = 13

y=9+3+1=13

10 y=x^2+2

= Parabola

Highest power is 2 → parabola (opens up, vertex (0,2))